

RAIDAINT WHEELS

Report submitted as part of DESIGN THINKING using AI (2501UC07)

BACHELOR OF TECHNOLOGY

G.Akash Reddy	25B11AI304
P.Leela Sri Ravi Kishore	25B11AI997
T.Jaswanth Reddy	25B11AIB87
S.Venkatesh	25B11AID23
CH.Yasaswini	25B11AI191
G.Akshaya	25B11AI355



ADB Road, Aditya Nagar, Surampalem,

Kakinada District -533437

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1. Introduction to Design Thinking

Design thinking is a strategic and practical process in which we use to tackle many complex problems that are ill-defined or unknown. This is because this process reframes these problems in a human-centric way. The process involves us to approach the user and understand their needs in an attempt to identify the solutions of the problems. It also offers us to think outside of the box in the effort of generating ground-breaking solutions. By using this process, we will make decisions based on what they want instead of only relying on assumptions and making risky bets. Based on the Hasso Plattner Institute of Design at Stanford (d. school). There are five phases of Design Thinking. The five phases are as follows:

- Emphasise – approach your users.
- Define – state your user’s need, problems, and your insights.
- Ideate – generating ideas based on problem-solving.
- Prototype – adopting a hands-on approach.
- Testing – solutions

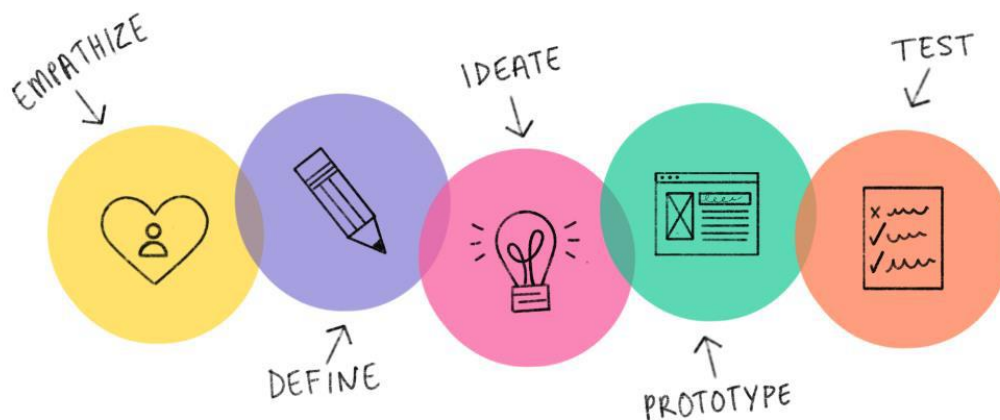


Figure 1: Phases of Design Thinking

PHASE 1 – Empathise

The first stage of Design Thinking is to empathise. This is where you are required to observe and engage yourself with the users to uncover their needs and then try to understand their problems. Empathy is very crucial in developing a human-centred process because it allows us as the thinkers to think logically and set aside our own assumptions.

PHASE 2 – Define

On this specific phase, we will analyse the information and observations that we have gathered in the previous phase. We will then try to synthesise all of them to define the core problems or to be more precise, a human-centred problem statement. This define phase will subsequently help us to gather great ideas to solve the problems or at the very least, it allows the users to resolve their issues with minimum difficulty.

PHASE 3 – Ideate

This is the phase where we start generating ideas following the previous phases. At this point, we have already had a solid background in generating brilliant ideas. The reason is that we have grown to understand our user's needs in the empathise phase, and created a problem statement in the define phase. There are many methods in ideation techniques such as Brainstorming. In this phase, we will need to come up with as many ideas as possible in developing the solutions and applying the concept of 'thinking outside of the box'.

PHASE 4 – Prototype

This is the phase where we turn the chosen ideas into a physical form of products as the solutions. This is also known as the experimental phase. The idea is that we need to find the best possible solution for each of the problems that have been identified in the previous phases. The prototypes are investigated, improved and re-examined. With this, they are either accepted or rejected based on how many problems that has been successfully tackled. At the end of this stage, we as the thinkers should be able to have a clearer view of the product and how the users should interact with the final product.

PHASE 5 – Testing

The last phase is testing. This is the phase where we try to get feedback on the chosen solutions from the users. This is a very important phase because it allows us as the thinker to identify as many flaws as possible as well as making improvements on the products to make it better. On the other hand, is also important to note that we can always return to the previous phases to make further refinements and alternative solutions.

2. Empathy

The first step in Design Thinking process is to empathise. Empathy is defined as the ability to understand the problems faced by client or the end-user, for that empathy is required to perceive yourself in client's way of thinking, to truly see the world through their eyes in a given context. Empathy is also the intersection of the designer and the user. As a designer, it is our job to find solutions or improve existing ways to design a system which resolve client's needs.

An Empathy map can be very useful in this step. It helps you obtain many of the required pieces of information that you will base your product on.

An interview with the end-user is often conducted during this process. A good interviewer needs to adopt these principles to get maximum data.

- Listen carefully to what client says, immerse yourself with the user.
- Engage with the user so that you understand their way of thinking.
- Understand what really the user does so that you get to know their preferences.
- Understand the feeling of the user in different contexts.
- Considers what emotions the user is experiencing at certain points.

Another way to access data is by observation - observe client's mannerism and body language or facial expressions to understand their feelings.

One of the main objectives of empathy is to identify user needs and behaviours that are latent, or unarticulated. In this process you should never assume anything or biased, design accordingly to user.

Understanding the Problem:

Every week, hundreds of hostel students hand over their clothes to the dhobi- and every week ,the same problem repeat .There's no system ,no tracking ,and no accountability .Students are left guessing when their clothes will return ,or worse ,whether they'll get the right clothes back at all.

Target Users

- Students living in a hostel who regularly give clothes for washing and face mix-up or delay issues.
- People staying in PGs who depend on shared laundry services.
- Workers who manage multiple customers and need an organized system to track clothes.
- Families or individuals using common laundry or local dhobi services.
- Authorities who want to improve service quality and reduce complaints.

Key User Needs

- Users need a way to ensure their clothes are not mixed or lost.
- Users want to know when their laundry will be ready without repeatedly asking.
- Users need an easy method (QR/token/SMS) to check laundry status.
- Users expect notifications or messages about collection and delivery.
- Users want confidence that their clothes are safe and properly handled.



Questionnaire and the answers

1. Do you face problems with laundry services?
sometimes clothes get mixed or delayed. It becomes difficult to track when they will be returned.

2. Would you prefer a tracking system for clothes?
It would make things easier and more organized.
I can know the status without asking repeatedly.

3. What is the biggest issue in current laundry services?
The main issue is lack of proper communication.
There is no clear update on delivery time.

4. Would you use a QR/token-based laundry system?
If it is simple and easy to use.
It will help avoid confusion and save time

3. Define

Problem Statement

In hostels and PG accommodations, laundry services are often managed manually, which leads to common issues such as clothes getting mixed up, delays in delivery, and lack of proper communication between users and dhobi workers. Students and residents frequently face confusion about the status of their clothes, resulting in inconvenience and loss of time. Therefore, there is a need for a simple and reliable system that can organize the laundry process and provide transparency to users.

Project Objective

The main objective of this project is to design and develop a smart laundry tracking system that improves the efficiency and reliability of dhobi services. This system aims to help users easily track their laundry status, receive timely updates, and avoid mix-ups or delays. By using simple technologies like QR codes or tokens and notifications, the project focuses on enhancing user convenience, ensuring better communication, and creating a more organized laundry management process.

4. Ideate

After the first two steps of empathizing and defining the problem, we continue along the process of Design Thinking to get to step three. This involves ideation or coming up with ideas that can help fulfill the requirements of the information gained in the first two steps. You can define a problem, keeping in mind all considerations, and you can also empathize with your subjects and their feelings, but those are futile if you cannot come up with the appropriate ideas to solve those issues. This makes the ideate phase very critical to the entire design thinking process.

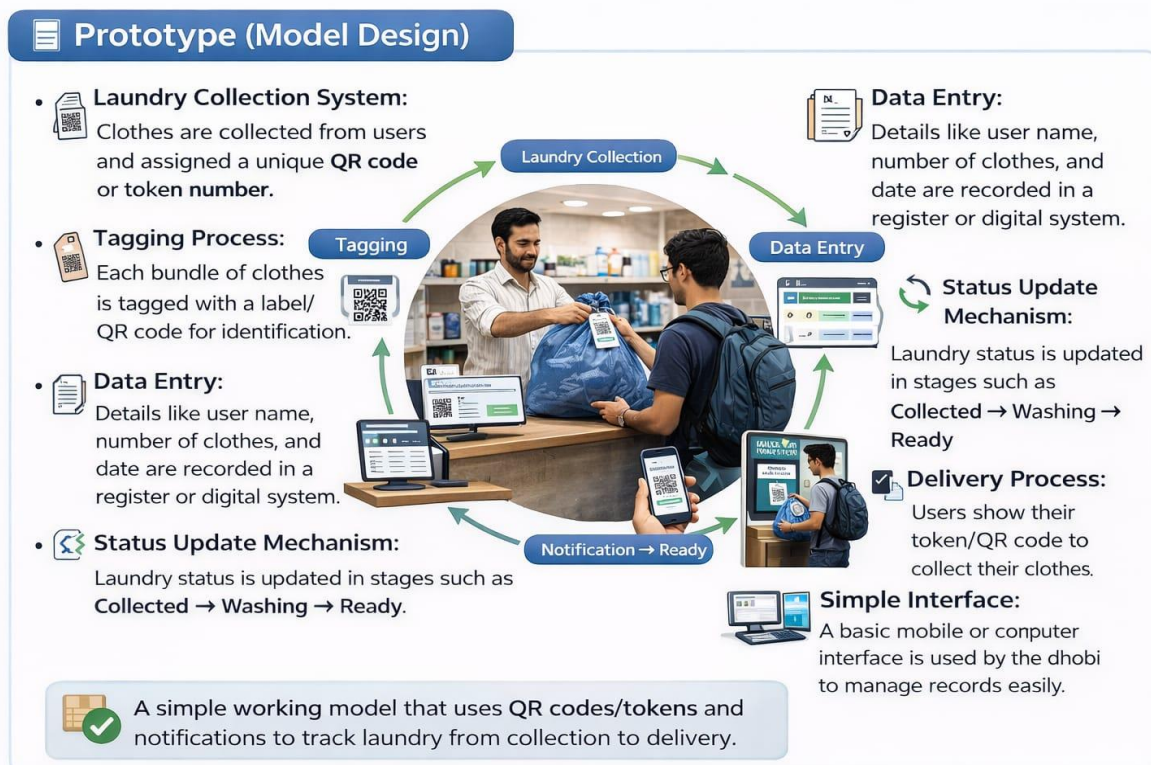
Concept development:

The concept of the Dhobi (Laundry) Tracking System is developed by focusing on simplifying and organizing the traditional laundry process using basic technology. The idea is to replace the manual method with a structured system that ensures proper tracking, communication, and delivery of clothes. The system is designed to be user-friendly so that both students and dhobi workers can use it without difficulty. It follows a step-by-step approach from collecting clothes to delivering them, ensuring transparency and reducing errors.

- Unique Identification: Each laundry item or bundle is assigned a QR code or token for easy tracking.
- Status Updates: The system provides updates like collected, washing, and ready for delivery.
- User Notification: Students receive SMS or app alerts when their clothes are ready.
- Data Management: All records are stored digitally to avoid confusion and loss of information.
- Ease of Use: The system is kept simple so that even non-technical users can operate it efficiently.

5. Prototype

- Laundry Collection System:**Clothes are collected from users and assigned a unique QR code or token number.
- Tagging Process:**Each bundle of clothes is tagged with a label/QR code for identification.
- Data Entry:**Details like user name, number of clothes, and date are recorded in a register or digital system.
- Status Update Mechanism:**Laundry status is updated in stages such as Collected → Washing → Ready.
- Notification System:**Users receive SMS or app notifications when their clothes are ready.
- Delivery Process:**Users show their token/QR code to collect their clothes safely.
- Simple Interface:**A basic mobile or computer interface is used by the dhobi to manage records easily.



6. Conclusion

The Dhobi Laundry Tracking System provides a simple and effective solution to the common problems faced in traditional laundry services, such as delays, misplacement of clothes, and lack of communication. By introducing basic technologies like QR codes or tokens along with notification systems, the process becomes more organized, transparent, and user-friendly. This project not only improves efficiency for dhobi workers but also enhances convenience and trust for users. Overall, it demonstrates how small innovations can create a big impact in solving everyday problems.

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